

The man behind it all is Dr. David Chaum. He formulated the blinded signature, a special form of a cryptographic signature that allowed a virtual coin to be signed without the signer seeing the actual coin, and permitted a form of digital money that offered anonymity and untraceability. This form of currency is known as Digital Cash.

Threshold cryptosystem

Threshold systems are designed to allow use only if a minimum number of parties, exceeding a threshold, agree to the said use. Technically, it means that in order to decrypt a ciphertext a minimum number of parties are required to collaborate in the process. Any less than that won't have sufficient information. For example, if in a bank at least 5 out of 10 people authorize the transaction, only then will it occur. Such systems obviate a single individual acting alone, while at the same time allowing many of the parties to be absent without the transaction being halted.

Most threshold cryptosystems have keys which are distributed into parts. The most common technique for partitioning a key is to form the key as the solution to equations in N variables. Only if all the N equations are known, the key can be determined by solving them. If any less than N equations are known, the key can't be determined since there's at least one independent variable in each equation. The minimum required threshold number can be chosen for N and the equations can be held by separate individuals. The same general concept can be used to form arbitrary combinations of key requirements by forming ORs and ANDs of encryptions using different sets of keys for different combinations of key holders. The major difficulties with such a system lie in the key distribution problem and the large number of keys necessary to achieve arbitrary key holder combinations.

Such systems are mostly employed in organizations with very valuable secrets, such as militaries and the governments. One of the applications is to store the secret information in multiple locations to prevent access to the ciphertext itself and thus prevent cryptanalysis on it.

Secure multi-party computation

Secure multi-party computation involves a set of parties with private inputs who wish to jointly compute a function of their inputs so that certain security properties (such as privacy and correctness) are preserved. It provides solutions to various real-life problems such as private auctions, distributed